INCH-POUND
MIL-DTL-24643/50D
22 August 2002
SUPERSEDING
MIL-C-24643/50C
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, FOUR CONDUCTOR, TYPE LSFNW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, uncoated, (see table I for size).
Second -	Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation or ethylene propylene rubber insulation (see table I for thickness). Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.
Fourth -	Four conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well-rounded assembly.
Fifth -	Binder tape applied helically with overlap.
Sixth -	Cross-linked polyolefin jacket. (see table I for thickness.)

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:15Z
Check the source to verify that this is the current version before use.

TABLE I. Details.

Military part no. M24643/50	Size	AWG conductor size	Insulation thickness min avg (in.)	Cable jacket thickness min avg (in.)	Overall diameter (in.) Type LSFNW	
					min.	max.
-01UN	3	16 (class B)	0.030	0.040	0.413	0.447
-02UN	4	14 (class B)	0.030	0.040	0.475	0.513
-03UN	9	10 (class B)	0.030	0.040	0.583	0.630
-04UN	23	7 (class B)	0.045	0.050	0.768	0.830

TABLE II. Electrical details.

Size 1/	AWG conductor size	Conductor resistance (ohms)	Insulation resistance (megohms)	Voltage withstand (minimum) (volts)	Cold bending mandrel diameter (max inch)	Accelerated service loading current (amperes)
3	16 (class B)	4.3	6000	4000	4	-----
4	14 (class B)	2.68	5000	4000	4	-----
9	10 (class B)	1.08	3800	4000	5	-----
23	7 (class B)	0.543	3700	5500	8	115

EXAMINATION AND TESTS:

Basic electrical: Requirements

- Conductor Resistance - ohms/1000 feet at 25°C, max ..... (see table II)
- Voltage withstand - volts, root mean square, minimum ..... (see table II)
- Insulation resistance –
  - Conductor to conductor, Megohms -1000 feet, minimum..... (see table II)
  - Conductor continuity ..... No failure

Group A:

- Visual and dimensional..... No failure

Group B:

- Cross-linked proof test (percent, maximum)
  - Insulation 50
  - Jacket (When tested at 200°C).....50
- Cold bending, cable - (see table I for mandrel diameter) ..... No damage
- Tear - pounds per inch thickness, minimum (ASTM D 470) 35

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:15Z
Check the source to verify that this is the current version before use.

Physicals (unaged)
Insulation
Ethylene propylene rubber
Tensile strength - lb/in^2, minimum 700
Elongation - percent, minimum 250
Cross-linked polyethylene
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

Group C:
Physicals - (aged)
Insulation
Ethylene propylene
Air oven
Tensile strength - percent of unaged, minimum 75
Elongation - percent of unaged minimum 75
Cross-linked polyethylene
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (conductor) - method 1 only, cycles, minimum 25
Permanence of printing (jacket) - cycles minimum.....125

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Requirements
Aging and compatibility (cable) (125 ±5°C) ..... No failure
Abrasion resistance (jacket) - scrapes, minimum 75

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:15Z
Check the source to verify that this is the current version before use.

Acid gas equivalent - percent, maximum
- Jacket 2
- Fillers 2
- Insulation 18

Halogen content - percent, maximum
- Jacket 0.2
- Fillers 0.2

Immersion (jacket)
- Tensile strength - percent of unaged, minimum 50
- Elongation - percent of unaged, minimum 50

Smoke index, maximum
- Jacket 25
- Fillers 45
- Insulation 45

Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum 80
Weathering (jacket) ..... No failure

UNIT ORDERING LENGTH:

Type and size	Feet (nominal)
LSFNW-3	2500
LSFNW-4, 9	2000
LSFNW-23	1500

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-048)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:15Z
Check the source to verify that this is the current version before use.